

Sheet 4

Question 1

A 1550 nm single mode digital fiber optic link needs to operate at 565 Mb/s over 50 km without repeaters. A single mode InGaAsP laser launches an average optical power of -13 dBm into the fiber. The fiber has a loss of 0.35 dB/km, and there is a splice with a loss of 0.1 dB every kilometer. The coupling loss at the receiver is 0.5 dB, and the receiver uses an InGaAs APD with a sensitivity of -39 dBm. Excess-noise penalties are predicted to be 1.5 dB. Set up an optical power budget for this link, find the system margin.

Question 2

Make a comparison and then calculate the maximum attenuation-limited transmission distance of the following two systems operating 100 Mb/s. (allow a system margin 6 dB for each system)

System 1	System 2
operates at 850 nm	operates at 1300 nm
GaAlAs laser diode: 0 dBm fiber coupled power	InGaAsP LED : -13 dBm fiber coupled power
Silicon avalanche photodiode : -50 dBm sensitivity	InGaAs pin photodiode: -38 dBm sensitivity
Graded index fiber : 3.5 dB/km attenuation at 850 nm	Graded index fiber : 1.5 dB/km attenuation at 1300 nm
Connector Loss 1 dB/connector	Connector Loss 1 dB/connector

Question 3

An engineer has the following components available

1. GaAlAs laser diode operates at 850 nm can couple 1mW into a fiber.
2. Ten sections of cable each of length 500 m with 4 dB/km attenuation and has connectors on both sides.
3. Connector loss 2dB/connector.
4. A *pin* photodiode receiver.
5. An avalanche photodiode receiver.

Using these components, the engineer wishes to construct a 5-km link operating at 20 Mb/s if the sensitivities of the pin and APD are -45 and -56 dBm respectively, which receiver can be used if a 6-dB system margin is required?

Question 4

It is required to design an optical communication link at the speed of 1 GB/Sec. RZ digital signal at the wavelength of 1500 nm between two cities separated by a distance of about 400 km. The optical components available have the following spec's at the operating wavelength:

- SMF with attenuation $\alpha = 0.3$ dB/Km and Dispersion $D = 2.5$ ps/(km.nm). The fiber is available in non-connectorized rolls, each one has a length of 25 km length.
- Optical Receiver with Sensitivity = -50 dbm @ 1500 nm and a Band Width BW = 3GHz (to achieve the required BER of 10^{-9})
- A Laser transmitter with an output power in the SMF = 1 mW, a rise time of 0.2 nsec., and a spectral width $\Delta\lambda = 1$ nm.

- All the components (except the fiber rolls) are terminated with connectors and the average connector loss is 0.5 dB per connection.
 1. Calculate the maximum distance “L” that can be achieved without repeater (the repeater_less distance) and thus the number of repeaters required for implementing the system.
 2. How many fiber rolls are required in the system.

Assume a system margin of **6 dB** is required in the system power budget and that the splicing loss is 0.2 dB.